

Entrega 3

Grupo1

Ejercicio 1

Métrica de Desempeño	TF Lite Micro (Framework)	Implementación C Puro	Análisis del Impacto Real
Tiempo de Inferencia (Δt)	~50 μs	~19 μs	Mejora del 62% en latencia al eliminar el overhead del intérprete.
Potencia Activa Sostenida (P)	~0.460 W (90 mA)	~0.450 W (88 mA)	Ligera reducción de ~10 mW en el pico de procesamiento de la CPU.
Energía por Inferencia ($E = P \cdot \Delta t$)	23.0 μJ (Microjulios)	8.55 μJ (Microjulios)	Reducción del 62.8% en consumo de energía real. El chip puede volver a dormir más rápido.
Precisión Matemática	Cuantizada (int8)	Punto Flotante (float32)	El C puro elimina el ruido de cuantización, siendo más exacto matemáticamente.
Flexibilidad/Escalabilidad	Alta (Soporta grafos complejos)	Nula (Requiere recodificar a mano)	TFLite es necesario para modelos de visión; C Puro es ideal solo para funciones fijas.

Uso de Memoria (RAM)	Alta (Intérprete + TensorArena)	Baja (Variables estáticas)	C puro es inmune a la fragmentación del heap dinámico.
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Las mediciones empíricas con el analizador USB demostraron que, si bien la potencia activa del microcontrolador a máxima carga varía levemente (0.46W en TF Lite vs 0.45W en C), la eficiencia real proviene del tiempo de ejecución. Al procesar la inferencia en 19 μ s en lugar de 50 μ s, la implementación en C puro consume solo 8.55 μ J de energía por operación frente a los 23.0 μ J del framework. Para un dispositivo alimentado por baterías que deba inferir miles de veces por minuto, esta reducción del 62% en la demanda energética justifica el esfuerzo de una implementación 'bare-metal' en modelos de baja dimensionalidad.

Fotos terminal de nuestra implementación:

```

HGLI@DESKTOP-HGLI082:~/Proyecto-Sumo-grupo-Juan-Lucas-Cristobal/Lab_3/projects_tf/hello_world$ idf.py -p /dev/ttyUSB0 monitor
Executing action: monitor
Running idf monitor in directory /home/diazb/Proyecto-Sumo-grupo-Juan-Lucas-Cristobal/Lab_3/projects_tf/hello_world
Executing "/home/diazb/esp/ldf-tools/python.env/ldf6.1.py3.12 env/bin/python /home/diazb/esp/ldf-tools/ldf_monitor.py -p /dev/ttyUSB0 -b 115200 --toolchain-prefix xtensa-esp323-elf- --target esp323 --revision 0 /home/diazb/Proyecto-Sumo-grupo-Juan-Lucas-Cristobal/Lab_3/projects_tf/hello_world/build/hello_world.elf -m /home/diazb/esp/ldf-tools/python.env/ldf6.1.py3.12 env/bin/python" /home/diazb/esp/ldf-tools/ldf.py "p" /dev/ttyUSB0 "...
... esp-ldf-monitor 1.9.0 on /dev/ttyUSB0 115200
... Quit: Ctrl+C | Menu: Ctrl+F | Help: Ctrl+F followed by Ctrl+H
ESP-ROM:esp323-20210327
Build:Mar 27 2021
py1 (0x1 (CODEDROM), boot:0x28 (SPI_FAST_FLASH_BOOT))
SPIWP:0xee
mode:DIO, clock div:1
load:0x3fc32200,len:0x1664
load:0x403c3700,len:0xf38
... 0x403c3700: sleep at 7?:?
load:0x403c3700,len:0x31ec
entry 0x403c3930
... 0x403c3930: call_start_cpu0 at /home/diazb/esp-ldf/components/bootloader/subproject/main/bootloader_start.c:227
(25) boot: ESP-IDF v6.1-dev-2038-g12f36a021f 2nd stage bootloader
(25) boot: compile time May 26 2026 16:38:31
(25) boot: Multicore bootloader
(27) boot: chip revision: v0.2
(28) boot: efuse block revision: v1.4
(33) via mode: Enabling QIO for flash chip W25Q128
(38) boot:esp323: Boot SPI Speed : 80MHz
(41) boot:esp323: SPI Mode : QIO
(45) boot:esp323: SPI Flash Size : 2MB
(49) boot: Enabling RNG early entropy source...
(53) boot: Partition Table:
(56) boot: ## Label Usage Type ST Offset Length
(62) boot: 0 nvs WiFi data 01 02 00000000 00000000
(69) boot: 1 phy_init RF data 01 01 00000000 00001000
(75) boot: 2 factory factory app 00 00 00010000 00100000
(82) boot: End of partition table
(85) esp_image: segment 0: paddr=00010020 vaddr=3c020020 size=0c3c4h ( 50116) map
(100) esp_image: segment 1: paddr=000113ec vaddr=3fc0f400 size=02f80h ( 12160) load
(103) esp_image: segment 2: paddr=00011374 vaddr=40374000 size=0bc44h ( 47264) load
(108) esp_image: segment 3: paddr=00020020 vaddr=42000020 size=1a014h ( 83988) map
(120) esp_image: segment 4: paddr=0003483c vaddr=4b374ca8 size=0a6d4h ( 42768) load
(136) esp_image: segment 5: paddr=0003cf10 vaddr=50000000 size=00024h ( 36) load
(142) boot: Loaded app from partition at offset 0x10000
(154) boot: Disabling RNG early entropy source...
(162) cpu_start: Multicore app
(162) cpu_start: GPIO 44 and 43 are used as console UART I/O pins
(163) cpu_start: Pre cpu start user code
(163) cpu_start: cpu freq: 160000000 Hz
(165) app_init: Application Information:
(169) app_init: Project name: hello_world
(173) app_init: App version: d8dc7862-dirty
(177) app_init: Compile time: May 26 2026 16:38:56
(182) app_init: ELF file SHA256: 547108d61...
(187) app_init: ESP-IDF: v6.1-dev-2038-g12f36a021f
(192) efuse_init: Max chip rev: v0.0
(196) efuse_init: Max chip rev: v0.99
(200) efuse_init: Chip rev: v0.2
(204) heap_init: Initializing RAM available for dynamic allocation:
(210) heap_init: AT 3FC93780 len 00005F00 (243 KiB): RAM
(215) heap_init: AT 3FC93710 len 00005724 (21 KiB): RAM
(220) heap_init: AT 3FCF0000 len 00000000 (0 KiB): DRAM
(226) heap_init: AT 600FE000 len 00001F08 (7 KiB): RTCRAM

```

```

E (234) spi_flash: flash io: qio
W (237) spi_flash: Detected size(16384k) larger than the size in the binary image header(2048k). Using the size in the binary image header.
I (250) sleep_gpio: Configure to isolate all GPIO pins in sleep state
I (256) sleep_gpio: Enable automatic switching of GPIO sleep configuration
I (263) main_task: Started on CPU0
I (283) main_task: Calling app_main()
x: 0.000 | y_TF: 0.000 (139 us) | y_Manual: 0.026 (62 us)
x: 0.314 | y_TF: 0.373 (60 us) | y_Manual: 0.291 (21 us)
x: 0.628 | y_TF: 0.559 (52 us) | y_Manual: 0.566 (19 us)
x: 0.942 | y_TF: 0.839 (51 us) | y_Manual: 0.819 (20 us)
x: 1.257 | y_TF: 0.966 (51 us) | y_Manual: 0.933 (20 us)
x: 1.571 | y_TF: 1.042 (51 us) | y_Manual: 0.996 (20 us)
x: 1.885 | y_TF: 0.957 (51 us) | y_Manual: 0.932 (20 us)
x: 2.199 | y_TF: 0.822 (51 us) | y_Manual: 0.810 (20 us)
x: 2.513 | y_TF: 0.534 (51 us) | y_Manual: 0.584 (20 us)
x: 2.827 | y_TF: 0.237 (51 us) | y_Manual: 0.289 (19 us)
x: 3.142 | y_TF: 0.008 (51 us) | y_Manual: -0.005 (19 us)
x: 3.456 | y_TF: -0.305 (51 us) | y_Manual: -0.299 (20 us)
x: 3.770 | y_TF: -0.534 (51 us) | y_Manual: -0.579 (20 us)
x: 4.084 | y_TF: -0.779 (51 us) | y_Manual: -0.839 (20 us)
x: 4.398 | y_TF: -0.966 (51 us) | y_Manual: -0.947 (20 us)
x: 4.712 | y_TF: -1.110 (51 us) | y_Manual: -1.006 (20 us)
x: 5.027 | y_TF: -0.983 (51 us) | y_Manual: -0.950 (20 us)
x: 5.341 | y_TF: -0.746 (51 us) | y_Manual: -0.809 (20 us)
x: 5.655 | y_TF: -0.534 (51 us) | y_Manual: -0.574 (20 us)
x: 5.969 | y_TF: -0.356 (51 us) | y_Manual: -0.307 (20 us)
x: 0.000 | y_TF: 0.000 (51 us) | y_Manual: 0.026 (20 us)
x: 0.314 | y_TF: 0.373 (51 us) | y_Manual: 0.291 (19 us)
x: 0.628 | y_TF: 0.559 (51 us) | y_Manual: 0.566 (19 us)
x: 0.942 | y_TF: 0.839 (51 us) | y_Manual: 0.819 (19 us)
x: 1.257 | y_TF: 0.966 (51 us) | y_Manual: 0.933 (19 us)
x: 1.571 | y_TF: 1.042 (51 us) | y_Manual: 0.996 (19 us)
x: 1.885 | y_TF: 0.957 (51 us) | y_Manual: 0.932 (19 us)
x: 2.199 | y_TF: 0.822 (51 us) | y_Manual: 0.810 (19 us)
x: 2.513 | y_TF: 0.534 (51 us) | y_Manual: 0.584 (19 us)
x: 2.827 | y_TF: 0.237 (51 us) | y_Manual: 0.289 (19 us)

```

diazb@DESKTOP-HGLI082:~/Proyecto-Sumo-grupo-Juan-Lucas-Cristobal/Lab_3/projects_tf/hello_world\$

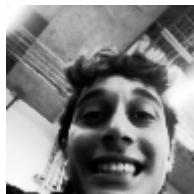
Fotos terminal ejemplo TFlite:

```
diazb@DESKTOP-HGLIQ82: ~/Proyecto-Sumo-grupo-Juan-Lucas-Cristobal/Lab_3/esp-tflite-micro/examples/hello_world
I (257) sleep_gpio: Enable automatic switching of GPIO sleep configuration
I (263) main_task: Started on CPU0
I (283) main_task: Calling app_main()
x_value: 0.000000, y_value: 0.000000
x_value: 0.314159, y_value: 0.372770
x_value: 0.628319, y_value: 0.559154
x_value: 0.942478, y_value: 0.838731
x_value: 1.256637, y_value: 0.965812
x_value: 1.570796, y_value: 1.042060
x_value: 1.884956, y_value: 0.957340
x_value: 2.199115, y_value: 0.821787
x_value: 2.513274, y_value: 0.533738
x_value: 2.827433, y_value: 0.237217
x_value: 3.141593, y_value: 0.008472
x_value: 3.455752, y_value: -0.304993
x_value: 3.769912, y_value: -0.533738
x_value: 4.084070, y_value: -0.779427
x_value: 4.398230, y_value: -0.965812
x_value: 4.712389, y_value: -1.109837
x_value: 5.026548, y_value: -0.982756
x_value: 5.340708, y_value: -0.745539
x_value: 5.654867, y_value: -0.533738
x_value: 5.969026, y_value: -0.355825
x_value: 0.000000, y_value: 0.000000
x_value: 0.314159, y_value: 0.372770
x_value: 0.628319, y_value: 0.559154
x_value: 0.942478, y_value: 0.838731
x_value: 1.256637, y_value: 0.965812
x_value: 1.570796, y_value: 1.042060
x_value: 1.884956, y_value: 0.957340
Done
diazb@DESKTOP-HGLIQ82: ~/Proyecto-Sumo-grupo-Juan-Lucas-Cristobal/Lab_3/esp-tflite-micro/examples/hello_world$
```

Ejercicio 2

2.1)

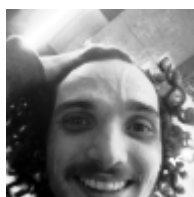
imagen 0



```
esp> detect_image 0

Total time = 555
FC time = 0
DC time = 114
conv time = 438
Pooling time = 0
add time = 0
mul time = 0
person score:66%, no person score 34%
I (67492) [esp_cli]: Time required for the inference is 558 ms
```

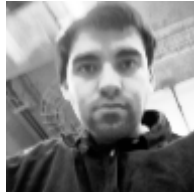
imagen 1



```
esp> detect_image 1

Total time = 555
FC time = 0
DC time = 115
conv time = 438
Pooling time = 0
add time = 0
mul time = 0
person score:94%, no person score 6%
I (71662) [esp_cli]: Time required for the inference is 559 ms
```

imagen 2



```
esp> detect_image 2

Total time = 555
FC time = 0
DC time = 114
conv time = 438
Pooling time = 0
add time = 0
mul time = 0
person score:95%, no person score 5%
I (74082) [esp_cli]: Time required for the inference is 558 ms
esp> |
```

2.2)

Link al drive con los videos y fotos: [📁 Lab 3](#)

Ejercicio 3

```
I (1354) TFLITE_MAIN: Initializing TFLite Micro Project...
I (1364) TFLITE_MAIN: Starting Batch Testing on 10 Static Images...
=====
I (2454) RESULT: Image 0 -> Time: 1073 ms | Raw Score: -32 | Confidence: 37.50%
I (3624) RESULT: Image 1 -> Time: 1073 ms | Raw Score: -128 | Confidence: 0.00%
I (4794) RESULT: Image 2 -> Time: 1073 ms | Raw Score: 127 | Confidence: 99.61%
I (5964) RESULT: Image 3 -> Time: 1073 ms | Raw Score: -124 | Confidence: 1.56%
I (7134) RESULT: Image 4 -> Time: 1073 ms | Raw Score: 126 | Confidence: 99.22%
I (8304) RESULT: Image 5 -> Time: 1073 ms | Raw Score: 106 | Confidence: 91.41%
I (9474) RESULT: Image 6 -> Time: 1073 ms | Raw Score: 127 | Confidence: 99.61%
I (10644) RESULT: Image 7 -> Time: 1073 ms | Raw Score: 126 | Confidence: 99.22%
I (11814) RESULT: Image 8 -> Time: 1073 ms | Raw Score: 127 | Confidence: 99.61%
I (12984) RESULT: Image 9 -> Time: 1073 ms | Raw Score: 127 | Confidence: 99.61%
=====
I (13084) TFLITE_MAIN: Batch Testing Completed.
I (13084) main_task: Returned from app_main()
```

Pares son con el identificador e Impares sin

Se detectó correctamente 6 de 10 imágenes, por lo que se tendrá que hacer nuevamente la IA para tener un mejor accuracy